

Evidence of Evolution Test Study Guide

1. What does evolution stand for? Evolution is the process where living things slowly change over time. These changes can help them survive better in their environment.
2. How is a homologous structure different from an analogous structure? Homologous structures: Same structure, different function (show common ancestor)- Analogous structures: Different structure, same function (no common ancestor)
3. How is a vestigial structure evidence for evolution? vestigial structures are body parts that no longer have a major use (like a human tailbone). They show that organisms changed over time.
4. What can the fossil record provide proof of evolution? The fossil record shows how organisms looked in the past and how they changed over time.

5. What information can a cladogram give about evolutionary relationships? A cladogram shows how species are related by evolution by displaying their common ancestors, shared traits, and how closely related they are.
6. How has molecular “DNA” testing changed what we know about evolutionary relationships? DNA testing lets scientists compare genes between species. The more similar the DNA, the more closely related the organisms are.
7. How are proteins involved in the process of electrophoresis? Proteins are separated by size using electricity. Smaller pieces move faster through the gel than larger ones.
8. Exactly how does DNA travel through the agar gel? DNA moves through the gel when electricity is applied. It travels toward the positive end because DNA is negatively charged.
9. What characteristics are used for a cladogram? Physical traits (bones, body parts) DNA similarities, Shared characteristics
10. What ways can embryology connect species to common ancestors? Embryology studies how organisms develop before birth. Similar embryos (like fish, birds, humans) suggest a common ancestor.

11. What information does the fossil record give us? What organisms lived long ago, When they lived, How they changed or went extinct

12. What is a half life? Half-life is the time it takes for half of a radioactive substance to decay.

13. How are fossils dated? Relative dating (layer position)-Radiometric dating (using isotopes)

14. What is an isotope and how is it used in dating fossils? An isotope is a form of an element with a different number of neutrons. Some isotopes decay over time and help date fossils.

15. What is the Law of Superposition? The oldest layer is on the bottom, The youngest layer is on top

16. What is speciation? Speciation is when one species splits into two new species over time.